ZentroMart AI Voice Support Agent

Evidence-grounded support | multi-turn workflows | Gnani.ai speech

REFERENCE DATE | 20 OCT 2026

OFFLINE TESTED

FULL EVALUATIONS PENDING

01 Problem and solution

ZentroMart support handles repeated questions about delivery, cancellations, returns, refunds, warranty and damaged goods. This capstone implements an AI support agent that retrieves company evidence, keeps conversation context, gates protected order lookups behind caller verification, and uses the same agent for text and voice.

02 Core capabilities

Evidence-grounded answers

Section-aware Markdown ingestion; source/effective-date metadata; policy precedence; abstention when evidence is absent.

Verified order support

Uses supplied lookup tool; asks for phone last four or delivery pincode; no order facts before tool verification.

Human handover

Uses the provided ticket tool for configured escalation rules; confirms a ticket only after `created=True`.

Voice interface

Browser microphone Record / Stop / preview / Send Audio, plus upload; audio is converted to WAV and sent through Gnani STT -> same agent -> Gnani TTS. Latest Windows logs show STT, TTS, API 200 and WAV fetch 206; explicitly confirm audible playback in the final demo.

03 Technology stack

Python | FastAPI | TF-IDF + cosine | OpenAI-compatible LLM | Gnani.ai STT / TTS

IMPLEMENTATION STATUS

45 automated tests pass, dataset validation passes, and retrieval finds 81/90 gold evidence IDs in the top five. The latest Windows voice run produced an STT transcript, Gemini response, completed Gnani TTS, `/api/voice` HTTP 200 and WAV fetch HTTP 206. Confirm audible playback during the final demo. Live baseline/RAG scoring and all 96 conversation turns remain pending unless rerun with configured credentials.

Architecture and implementation

One shared agent for text and voice

ONE TURN / END-TO-END PIPELINE

DATA PLANE 21 KB Markdown documents + 70 product rows | 7 database CSVs | holidays.csv | supplied `lookup_order` and `create_ticket` tools | practice questions and conversations

04 Implementation details

Ingestion and chunking

Reads front matter, splits on `[SECTION-ID]` headings and paragraph boundaries, and preserves source, status, effective date, version and chunk index.

Retrieval and authority

TF-IDF n-gram vectors + cosine similarity; current policies receive priority and a per-source diversity cap. Superseded return text is conditional-only for the legacy case preserved by current RET-3.

Session and protected data

Session state retains recent turns and order context. The supplied tool enforces proof matching; contact fields and pincode proof are filtered from model context.

Escalation and tool truth

Order lookup, business-day checks and ticket actions are deterministic code. Ticket ID/handover wording depends on the actual tool result; failed creation is reported as unconfirmed.

SPEECH INTEGRATION BOUNDARY

Official `gnani-vachana` adapters are configured for STT `en-IN` and TTS `timbre-v2.5` / `Kaveri`. Browser `MediaRecorder` output is decoded and encoded as true WAV before upload. No alternate speech provider is used. The owner-reported Windows run completed STT, response generation, TTS and WAV retrieval through the microphone workflow. Explicit audible-playback confirmation remains a final demo check; no live provider credentials are stored in this build workspace.

Evaluation and verified results

Offline metrics only where measured

45 / 45
pytest tests passed

90.0%
gold evidence Recall@5

47 / 47
questions hit at least one gold item

0
fabricated scores claimed

RESULTS / STATUS

Metric	Measured result	Interpretation
Dataset validator	PASS	9 CSVs, 21 KB docs, keys, order math, practice evidence and verification gates checked.
Automated test suite	45 passed / 0 failed	pytest run with offline mocks and supplied-data integration tests.
Gold evidence Recall@5	81/90 = 90.0%	Exact gold section/source IDs found in top-five retrieved chunks.
Question hit rate@5	100% (47/47)	At least one gold evidence item retrieved for each question with a gold label.
Step 1 baseline accuracy	PENDING	Requires configured live LLM; no answer accuracy inferred from retrieval alone.
Step 2 RAG answer accuracy	PENDING	Same 52 questions and rubric await the live LLM evaluation.
Conversation evaluation	0 / 96 turns executed	Live evaluation status; answer and context checks require an LLM-configured run.
Gnani STT / TTS	WINDOWS MIC RUN OBSERVED	Latest owner-reported logs show STT/TTS complete, `/api/voice` 200 and WAV fetch 206; confirm audible playback on final demo.
Application / batch smoke	PASS (offline + manual)	Offline smoke covered `/` and `/health`; owner logs also show live `/api/chat` and `/api/voice` 200 on Windows. Full scoring evaluations remain pending.

HOW TO READ THE RETRIEVAL SCORE

Recall@5 = the fraction of supplied gold evidence IDs matched against the top-five retrieved source/section metadata. The 100% question hit rate means at least one gold item was found for each of 47 labelled questions. Neither metric proves the generated answer is correct. Product queries with unspecified variants can remain ambiguous; baseline/RAG correctness, unsupported-answer rate and conversation pass rate remain unmeasured.

Limitations and reproduction

Remaining steps before final submission

05 Limitations and next work

Full scores and final playback confirmation pending.

The owner-reported Windows microphone run reached Gnani STT, Gemini, TTS and returned the WAV. Explicitly confirm audible playback on the final demo; run baseline/RAG and 96-turn evaluations with configured provider credentials.

Lexical retrieval, not dense embeddings.

Current retriever is sparse TF-IDF with cosine similarity and intent-aware ranking. Evaluate neural embeddings or hybrid reranking if allowed, using the same gold evidence set.

Session state is in memory.

Restarting the app loses context; use durable, access-controlled session storage and stronger identity controls before production.

Demo and public repo pending.

No fabricated `demo.mp4` was added. Record a real <=120-second demo after direct-microphone/live provider checks; publish this local project to a public repo and verify access.

06 Fixes made

The current build adds browser microphone recording with WAV conversion, validated voice uploads, ticket CSV readback and lookup-first routing to avoid duplicate ticket creation. Regression tests cover these paths. Dataset validation, order verification and honest tool-result handling remain exercised by tests.

07 Reproduce locally

COMMANDS (FROM PROJECT ROOT)

```
pip install -r requirements.txt

# copy .env.example to .env

# set LLM_API_KEY, LLM_MODEL,

# GNANI_API_KEY, DATASET_DIR

python scripts/build_index.py

DATASET_DIR=<dataset-root> pytest -q

python scripts/evaluate_questions.py

python scripts/evaluate_conversations.py

uvicorn app:app --host 127.0.0.1 --port 8000

python run.py --input-dir sample_audio \

  --output-dir outputs
```

Links / submission status

GitHub: github.com/aayush-as69/voiceops-zentromart
Verify this source is pushed and public before review.
Demo MP4: pending final audible-playback check and recording.
SDK: github.com/Gnani-AI-Mintlify/gnani-vachana

Batch command

`python run.py --input-dir sample_audio --output-dir outputs` processes actual audio files; missing configuration exits clearly without mock substitutions.